

These are notes for Malek Abdesselam's Functional Analysis class for Fall 2026.

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Motivation

Consider a space V of (sufficiently nice) functions $f: \mathbb{R} \rightarrow \mathbb{R}$. Then, similar to how a vector in $\mathbb{R}^n$ can be denoted as

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may view the function $f: \mathbb{R} \rightarrow \mathbb{R}$ as a “column vector” with uncountable index. That is, $f = (f(x))_{x \in \mathbb{R}}$. If $L: V \rightarrow V$ is a linear map, then an analogue of the finite-dimensional matrix representation can be viewed as a kernel that f is integrated against; that is,

$$(Lf)(x) = \int_{\mathbb{R}} K(x, y) f(y) dy.$$

In the special case that $L = \text{id}_V$, then the kernel $K(x, y)$ is given by the Dirac delta distribution, $K(x, y) = \delta(x - y)$. The space of these distributions is the primary object of study in Fourier analysis.

Notation

- The natural numbers are given by $\mathbb{N} = \{0, 1, 2, 3, \dots\}$.
- The strictly positive natural numbers are given by $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$.
- A multi-index is a tuple $\alpha \in \mathbb{N}^n$. We let

$$|\alpha| = \alpha_1 + \dots + \alpha_n.$$

If $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, then we will write

$$x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$$

and

$$\alpha! = \alpha_1! \cdots \alpha_n!$$

for the corresponding values.

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is any function, then we will write

$$\partial^\alpha f = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

Operations and comparators on multi-indices will be taken componentwise.

- If A is a set, we will write $\mathcal{P}(A)$ for the power set of A . If A and B are sets, then the set $B^A \subseteq \mathcal{P}(A \times B)$ is the set of all functions $f: A \rightarrow B$.

We will write $\mathcal{P}_{\text{fin}}(X)$ to consist of all finite subsets of X .

- If $x \in \mathbb{R}^n$ is given by $(x_1, \dots, x_n)$, we will define the bracket

$$\begin{aligned} \langle x \rangle &:= \sqrt{1 + x_1^2 + \cdots + x_n^2} \\ &= \sqrt{1 + |x|^2}. \end{aligned}$$

In the particular case of $n = 1$, we will write $\langle k \rangle = \sqrt{1 + k^2}$ for any $k \in \mathbb{N}$. Note here that $|\cdot|$ denotes the norm on Euclidean space.

- If $n \in \mathbb{N}$, then

$$[n] = \{1, 2, \dots, n\},$$

where we have $[0] = \emptyset$.

- If A is a finite set, then $|A|$ is the cardinality of A .
- Generally we will denote by $\mathbb{K}$ for either $\mathbb{R}$ or $\mathbb{C}$.

Locally Convex Topological Vector Spaces

Recall that if X is a set, then $\mathcal{T} \subseteq \mathcal{P}(\mathcal{P}(X))$ is a topology on X if it satisfies:

- $\emptyset, X \in \mathcal{T}$;
- $U_1, U_2 \in \mathcal{T}$ implies $U_1 \cap U_2 \in \mathcal{T}$;
- $\{U_i : i \in I\} \subseteq \mathcal{T}$ implies $\bigcup_{i \in I} U_i \in \mathcal{T}$.

Definition: Definitions

We say $(V, \mathcal{T})$ is a *topological vector space* (over $\mathbb{K}$) if V is a $\mathbb{K}$ -vector space whose topology is such that the maps

$$\begin{aligned} V \times V &\rightarrow V \\ (u, v) &\mapsto u + v \end{aligned}$$

and

$$\begin{aligned} \mathbb{K} \times V &\rightarrow V \\ (\lambda, v) &\mapsto \lambda v \end{aligned}$$

are continuous with respect to the product topology. Note that $\mathbb{K}$ is equipped with its traditional topology.

The morphisms of topological vector spaces are the *continuous* linear maps. We say $V \cong W$ are isomorphic as topological vector spaces if there is $\varphi: V \rightarrow W$ that is bijective, linear, and continuous, and φ^{-1} is also continuous.

Definition: Let V be a $\mathbb{K}$ -vector space. We say $\rho: V \rightarrow [0, \infty)$ is a *seminorm* if

$$\begin{aligned}\rho(v + w) &\leq \rho(v) + \rho(w) \\ \rho(\lambda v) &= |\lambda| \rho(v)\end{aligned}$$

for any $v, w \in V$ and $\lambda \in \mathbb{K}$.

Remark: The only difference between a seminorm and a norm is that $\rho(v) = 0$ need not imply that $v = 0$. A norm is a seminorm.

We will let $\text{SN}(V)$ be the set of all seminorms of V .

Definition: Let V be a $\mathbb{K}$ -vector space. Let $\mathcal{N} \subseteq \text{SN}(V)$. Define $\mathcal{T}(\mathcal{N})$ to be a topology with basic open sets $U(x, F, \varepsilon) \subseteq V$ given by

$$\begin{aligned}U(x, F, \varepsilon) &= \{y \in V : \rho(y - x) < \varepsilon \text{ for all } \rho \in F\} \\ &= \bigcap_{\rho \in F} U(x, \rho, \varepsilon).\end{aligned}$$

for any $x \in V$, $F \subseteq \mathcal{N}$ finite, and $\varepsilon > 0$.

Remark: I prefer to use the subbasic open sets

$$U(x, \rho, \varepsilon) = \{y \in V : \rho(y - x) < \varepsilon\}$$

and write intersections.

We observe that the subsets of the form $U(x, \{\rho\}, \varepsilon)$ as ρ ranges across $\mathcal{N}$ forms a subbase for the topology of V .

Proposition: If $F \subseteq \mathcal{N}$ is finite, $\varepsilon > 0$, and $x \in V$, then

$$U(x, F, \varepsilon) \in \mathcal{T}(\mathcal{N}).$$

Proof. Suppose $y \in U(x, F, \varepsilon)$. Our goal is to find $\eta > 0$ such that $U(y, F, \eta) \subseteq U(x, F, \varepsilon)$. Suppose $z \in U(y, F, \eta)$. If $\rho \in F$, then

$$\begin{aligned}\rho(z - x) &\leq \rho(z - y) + \rho(y - x) \\ &< \eta + \rho(y - x).\end{aligned}$$

Therefore, we will define η to be small enough such that $\eta + \max_{\rho \in F} \rho(y - x) < \varepsilon$. This gives the desired result. $\square$

Definition: A topological space is called a locally convex topological vector space (locally convex topological vector space) if the topology is generated by a family of seminorms $\mathcal{N}$ with the property that

$$\bigcap_{\rho \in \mathcal{N}} \{v : \rho(v) = 0\} = \{0\}.$$

A family $\mathcal{N}$ of seminorms that satisfies this property is said to be *separating*.

Example: If H is a Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$, then the seminorms $v \mapsto |\langle v, e_i \rangle|$ are separating.

Theorem: If V is a $\mathbb{K}$ -locally convex topological vector space and $\mathcal{N}$ is a collection of seminorms that defines the topology on V , then the pair $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff if and only if $\mathcal{N}$ separates the points of V .

Proof. Assume $\mathcal{N}$ separates the points of V . Suppose $v \neq w$ are distinct elements of V . Then, since $v - w \neq 0$, there is some $\rho \in \mathcal{N}$ such that $\rho(v - w) = \varepsilon_0 > 0$. Consider the finite set $F = \{\rho\}$, and choose some $0 <$

$\varepsilon < \varepsilon_0/2$. Then, we claim that the basic open sets $U(v, F, \varepsilon)$ and $U(w, F, \varepsilon)$ are disjoint. This follows from the fact that if $x \in U(v, F, \varepsilon)$, then $\varepsilon_0 = \rho(v - w) \leq \rho(v - x) + \rho(x - w)$, so that $\rho(w - x) \geq \varepsilon_0 - \varepsilon > \varepsilon_0 - \varepsilon_0/2 > \varepsilon$, yielding the desired result.

Now, suppose $\mathcal{N}$ does not separate the points of V . Then, there are some distinct $v \neq w$ such that for all $\rho \in \mathcal{N}$, we have $\rho(v - w) = 0$. In particular, observe that for all finite subsets $F \subseteq \mathcal{N}$ and all $\varepsilon > 0$, we have that w is contained in $U(v, F, \varepsilon)$, meaning that there are no basic open sets that separate v and w . Since any open subset of V contains a basic open subset, it follows that for any open subsets U and U' , we have some finite F such that $U(v, F, \varepsilon) \subseteq U$, but this also has $w \in U(v, F, \varepsilon)$, implying that $U \cap U' \neq \emptyset$. Thus, V is not Hausdorff. $\square$

Examples

Remark: If we remove the assumption that the family $\mathcal{N}$ is separating, then the topology induced by the trivial seminorm $\rho: V \rightarrow [0, \infty)$ is the indiscrete topology on V . Similarly, there is a family of seminorms that induces the discrete topology on V .

- (i) Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Examples of these include Banach spaces such as $L^p(X, \Sigma, \mu)$, where Σ is a σ -algebra and μ is a positive measure on the measurable space (X, Σ) , and $p \in [1, \infty]$.

There is a metric on this normed space $d(x, y) = \|x - y\|$, which gives rise to the metric topology $\mathcal{T}(d)$. We have $U \in \mathcal{T}(d)$ if and only if $U \subseteq V$ is such that for any $x \in U$, there exists $\varepsilon > 0$ such that

$$\begin{aligned} \{y \in V : d(x, y) < \varepsilon\} &= U(x, \varepsilon) \\ &\subseteq U. \end{aligned}$$

This is a locally convex space with seminorm $\mathcal{N} = \{\|\cdot\|\}$. There are two possibilities for finite subsets $F \subseteq \{\|\cdot\|\}$, meaning that if $F = \emptyset$, then $U(x, F, \varepsilon) = U(x, \emptyset, \varepsilon) = V$, while if $F = \{\|\cdot\|\}$, then

$$\begin{aligned} U(x, F, \varepsilon) &= \{y \in V : \rho(x - y) < \varepsilon \text{ for all } \rho \in F\} \\ &= \{y \in V : \|x - y\| < \varepsilon\} \\ &= U(x, \varepsilon). \end{aligned}$$

- (ii) Define $s(\mathbb{N}, \mathbb{K})$ to be the space of sequences $x = (x_n)_{n \geq 0}$ in $\mathbb{K}^{\mathbb{N}}$ such that for all $k \geq 0$, $\sup_{n \geq 0} \langle n \rangle^k |x_n| < \infty$, where $\langle n \rangle = \sqrt{1 + n^2}$ is the bracket. This is the space of *rapidly decaying sequences* in $\mathbb{K}$. In essence, elements of $s(\mathbb{N}, \mathbb{K})$ decay faster than any inverse power of the index n .

We define the seminorms

$$\rho_k(x) = \sup_{n \geq 0} \langle n \rangle^k |x_n|.$$

In fact, the $\{\rho_k : k \in \mathbb{N}\}$ are norms. Let

$$\mathcal{N} := \{\rho_k : k \in \mathbb{N}\}$$

We obtain a locally convex on $s(\mathbb{N}, \mathbb{K})$ with topology induced by the family $\mathcal{N}$.

Later, we will see that $(s(\mathbb{N}, \mathbb{K}), \mathcal{T}(\mathcal{N}))$ is *not* a normed space. That is, there does not exist a norm $\|\cdot\|$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\{\|\cdot\|\})$.

- (iii) Take $V = C([0, \infty), \mathbb{R})$ to be the set of all real-valued continuous functions on $[0, \infty)$ with pointwise addition and scalar multiplication. This is a real vector space.

At first, we may think $\|f\| = \sup_{t \in \mathbb{R}} |f(t)|$ would define a norm, but this is not satisfactory since the functions in V are not necessarily bounded.

Therefore, we want to restrict to compact subsets. Take

$$\rho_k(f) := \sup_{0 \leq t \leq k} |f(t)|.$$

Notice that the ρ_k are not norms (use Urysohn's Lemma or your favorite piecewise function to construct one). However, the family $\mathcal{N} = \{\rho_k : k \geq 0\}$ is a separating family.

Consider the set of functions $V_0 = \{f \in V : f(0) = 0\}$. Then, $V_0 \leq V$ is a closed subset.

Proof. Consider the linear functional $\text{ev}_0 : V \rightarrow \mathbb{R}$ given by $f \mapsto f(0)$. We claim that this linear functional is continuous.

Let $U \subseteq \mathbb{R}$ be open. Let $z \in U$, and let $\varepsilon > 0$ be such that $U(z, \varepsilon) := \{y \in \mathbb{R} : |z - y| < \varepsilon\}$ is contained in U . Let $f \in \text{ev}_0^{-1}(U(z, \varepsilon))$, and set $\varepsilon_0 = |z - f(0)|$. To show that $\text{ev}_0^{-1}(U(z, \varepsilon))$ is open, it suffices to find some seminorm ρ_k and some $\delta > 0$ such that $U(f, \rho_k, \delta) \subseteq \text{ev}_0^{-1}(U(z, \varepsilon))$.

In fact, we may choose $k = 1$ and $\delta = \varepsilon - \varepsilon_0$. Suppose $g \in U(f, \rho_1, \delta)$. Then, we have that

$$\begin{aligned} |g(0) - z| &\leq |g(0) - f(0)| + |f(0) - z| \\ &\leq \sup_{t \in [0, 1]} |g(t) - f(t)| + \varepsilon_0 \\ &< \delta + \varepsilon_0 \\ &= \varepsilon. \end{aligned}$$

In particular, Thus, $U(f, \rho_1, \delta) \subseteq \text{ev}_0^{-1}(U(z, \varepsilon))$, so that $\text{ev}_0^{-1}(U(z, \varepsilon))$ is open as any element in this set admits a basic open neighborhood contained in the set. Therefore, ev_0 is continuous.

Since $V_0 = \text{ev}_0^{-1}(\{0\})$, it follows that V_0 is closed since $\{0\} \subseteq \mathbb{R}$ is closed. $\square$

We claim that the subspace topology on V_0 is given by $\mathcal{T}_0 = \mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$.

Proof. First, we will show that, given a subbasic open neighborhood $U = V_0 \cap U(x, \rho_k, \varepsilon)$ in the subspace topology of V_0 and any element y in this subbasic open neighborhood, there is an open neighborhood $V \in \mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$ such that $y \in V \subseteq U$.

If $y \in U$, we observe that $y \in U(x, \rho_k, \varepsilon)$. In particular, since these form a subbase for $\mathcal{T}(\{\rho_k : k \geq 0\})$, it follows that there is some finite collection $F = \{\rho_{i_j} : j = 1, \dots, m\}$ of seminorms and some $\delta > 0$ such that

$$U(y, F, \delta) = \bigcap_{j=1}^m U(y, \rho_{i_j}, \delta).$$

Since $U(y, \rho_{i_j}|_{V_0}, \delta) \subseteq U(y, \rho_{i_j}, \delta)$ for each j , it follows that we have

$$\bigcap_{j=1}^m U(y, \rho_{i_j}|_{V_0}, \delta) \subseteq V_0 \cap \left(\bigcap_{j=1}^m U(y, \rho_{i_j}, \delta) \right),$$

whence setting V to equal the left hand side yields the desired result. Therefore, $\mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$ refines the subspace topology $\mathcal{T}_0$.

In the reverse direction, observe that if $y \in U(x, \rho_j|_{V_0}, \varepsilon)$ in V_0 , then $y \in U(x, \rho_j, \varepsilon)$, so there is some $\delta > 0$ and finite $F = \{\rho_{i_j} : j = 1, \dots, m\}$ such that

$$U(y, F, \varepsilon) = \bigcap_{j=1}^m U(y, \rho_{i_j}, \delta)$$

$$\subseteq U(x, \rho_j, \varepsilon),$$

but we notice then that $U(x, \rho_j, \varepsilon) \cap V_0 \subseteq U(x, \rho_j|_{V_0}, \varepsilon)$, so that $V_0 \cap U(y, F, \varepsilon) \subseteq U(x, \rho_j|_{V_0}, \varepsilon)$, so that $\mathcal{T}_0$ refines $\mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$. Thus, these topologies are equal. $\square$

We may consider $\mathcal{B}(\mathcal{T}_0)$ to be the Borel σ -algebra on V_0 . Brownian motions are a special class of probability measure on $(V_0, \mathcal{B}(\mathcal{T}_0))$. Specifically, they are the probability measures such that for every $n \geq 0$, partition $0 < t_1 < t_2 < \dots < t_n$, the random variables

$$\begin{aligned} X_1 &= f \mapsto f(t_1) \\ X_2 &= f \mapsto (f(t_2) - f(t_1)) \\ X_3 &= f \mapsto (f(t_3) - f(t_2)) \\ &\vdots \\ X_n &= f \mapsto (f(t_n) - f(t_{n-1})) \end{aligned}$$

are independent centered Gaussian random variables with variances $t_1, t_2 - t_1, \dots, t_n - t_{n-1}$ for an arbitrary f .

Similar to the case of $s(\mathbb{N}, \mathbb{K})$, the space $(V_0, \mathcal{T}_0)$ is not a normed space.

- (iv) Consider the vector space $V = \mathbb{R}^{\mathbb{N}}$, which is the space of all real-valued sequences. Let $\mathcal{T}$ be the product topology on $\mathbb{R}^{\mathbb{N}}$. We are curious as to whether $(V, \mathcal{T})$ is a locally convex topological vector space.

Consider the function $d: V \times V \rightarrow [0, \infty)$ defined by

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min(1, |x_n - y_n|).$$

Then, we have that d is a distance metric, and $\mathcal{T} = \mathcal{T}(d)$.

Proof. We start by showing that if $x \in V$ is any element and $\varepsilon > 0$, then $U(x, \varepsilon)$ is open in $\mathcal{T}$. Translating, we take $x = 0$ (since the metric d is translation-invariant), so that we are now seeking to show that $U(0, \varepsilon)$ is open. This is equivalent to showing that if $x \in U(0, \varepsilon)$, then there is a finite collection of subbasic open sets $\pi_j^{-1}(U_j)$ whose intersection W contains x and is contained in $U(0, \varepsilon)$.

To start, we observe that there is some index n_0 such that

$$\sum_{n=n_0+1}^{\infty} 2^{-n} < \varepsilon/2.$$

We then select $U_j = (x_j - \varepsilon/4, x_j + \varepsilon/4)$ for each $j = 0, \dots, n_0$. It then follows that $x \in W$, and we have for any $y \in W$ that

$$\begin{aligned} d(x, y) &= \sum_{j=0}^{\infty} 2^{-j} \min(1, |x_j - y_j|) \\ &\leq \sum_{j=0}^{n_0} 2^{-j} \min(1, |x_j - y_j|) + \sum_{j=n_0+1}^{\infty} 2^{-j} \\ &< \sum_{j=0}^{n_0} 2^{-j} (\varepsilon/4) + \varepsilon/2 \\ &\leq \varepsilon, \end{aligned}$$

whence $x \in W \subseteq U(0, \varepsilon)$.

Now, consider a basic open set $W = \bigcap_{j=0}^{n_0} \pi_j^{-1}(U_j)$. Let $x \in W$. Let $\delta_0, \dots, \delta_{n_0}$ be such that $(x_j - \delta_j, x_j + \delta_j) \subseteq U_j$ for each $j = 1, \dots, n$, and set $\delta = 2^{-n_1} \min(\delta_0, \dots, \delta_{n_0})$, where n_1 is any integer strictly larger than n_0 .

Notice then that if $y \in U(x, \delta)$, then we have for each $j = 0, \dots, n_0$ that

$$\begin{aligned} |x_j - y_j| &< 2^j \delta \\ &\leq 2^{n_0} 2^{-n_1} \min(\delta_0, \dots, \delta_{n_0}) \\ &< \min(\delta_0, \dots, \delta_{n_0}) \\ &\leq \delta_j. \end{aligned}$$

In particular, this means we must have $U(x, \delta) \subseteq W$.

Therefore, we have that $\mathcal{T}$ and $\mathcal{T}(d)$ generate the same topology on V . □

In particular, we may consider the seminorms

$$\rho_n(x) = |x_n|$$

and $\mathcal{N} = \{\rho_n : n \geq 0\}$. Then, we claim that $\mathcal{T} = \mathcal{T}(\mathcal{N})$.

Proof. A basic open neighborhood of $x \in V$ in $\mathcal{T}(\mathcal{N})$ is given by the sets

$$U(x, F, \varepsilon) = \bigcap_{j \in F} \{(y_n)_n : |x_j - y_j| < \varepsilon\}.$$

Meanwhile, a basic open neighborhood of x in $\mathcal{T}$ is given by the sets

$$U'(x, F, \varepsilon) = \bigcap_{j \in F} \pi_j^{-1}((x_j - \varepsilon, x_j + \varepsilon)).$$

In fact, these are the same subsets of V by the definition of the projection map $\pi_j : V \rightarrow \mathbb{R}$ taking $(x_n)_n \mapsto x_j$.

Therefore, if we have $y \in U(x, F, \varepsilon)$ in $\mathcal{T}(\mathcal{N})$, then if we define

$$\delta = \frac{1}{2} \min(\{|x_j - y_j| : y_j \in F\}),$$

we have that $U'(y, F, \delta) \subseteq U(x, F, \varepsilon)$. Similarly, if $y \in U'(x, F, \varepsilon)$, then setting δ to equal the same as above, we have $U(y, F, \varepsilon) \subseteq U'(x, F, \varepsilon)$.

This gives $\mathcal{T} = \mathcal{T}(\mathcal{N})$. □

In particular, V is not a normed space.

- (v) Consider $c_{00}(\mathbb{N}, \mathbb{K}) \cong \mathbb{K}[x]$ to consist of all the finitely supported sequences $(x_n)_{n \geq 0}$. This is a vector space. Consider the seminorms $\mathcal{N} = \text{SN}(V)$, and the locally convex space $(V, \mathcal{T}(\mathcal{N}))$. Then, this is a locally convex space that is not metrizable. The topology $\mathcal{T}(\mathcal{N})$ is known as the finest locally convex topology.

General Properties of Topological Vector Spaces

Definition: Let $(V, \mathcal{T})$ be a $\mathbb{K}$ -TVS, and let $\tau \in \text{SN}(V)$. Then, τ is a *continuous seminorm* for $(V, \mathcal{T})$ if $\tau : V \rightarrow [0, \infty) \subseteq \mathbb{R}$ is continuous.

Proposition: Let V be a $\mathbb{K}$ -vector space, $\mathcal{N} \subseteq \text{SN}(V)$ define a locally convex topology on V . If $\tau \in \text{SN}(V)$, then τ is continuous for $(V, \mathcal{T}(\mathcal{N}))$ if and only if there exists $C \geq 0$, $n \in \mathbb{N}$, and $\rho_1, \dots, \rho_n \in \mathcal{N}$ such that

$$\tau(v) \leq C \max\{\rho_j(v) : j = 1, \dots, n\} \quad (*)$$

or

$$\tau(v) \leq C \sum_{j=1}^n \rho_j(v).$$

Proof. Suppose τ is continuous. Then, $[0, 1)$ is open in $[0, \infty)$, so $\tau^{-1}([0, 1))$ is open in $\mathcal{T}(\mathcal{N})$ with $0 \in \tau^{-1}([0, 1))$.

So, by definition, there exists $\varepsilon > 0$ and $\rho_1, \dots, \rho_n$ such that

$$\bigcap_{j=1}^n U(0, \rho_j, \varepsilon) \subseteq \tau^{-1}([0, \infty)).$$

That is, for any $i \in [n]$, we have that if $\rho_i(x) < \varepsilon$, then $\tau(x) < 1$.

Let $x \in V$ and $\lambda > 0$. Then,

$$\begin{aligned} \tau(x) &= \tau(\lambda \lambda^{-1} x) \\ &= \lambda \tau(\lambda^{-1} x) \\ &< \lambda \end{aligned}$$

if for every i , we have $\rho_i(\lambda^{-1} x) < \varepsilon$.

Define $J = \{\lambda > 0 : \rho_i(\lambda^{-1} x) < \varepsilon \text{ for all } i \in [n]\}$. Rewriting, we have

$$\begin{aligned} J &= \left\{ \lambda > 0 : \frac{1}{\varepsilon} \rho_i(x) < \lambda \text{ for all } i \in [n] \right\} \\ &= \bigcap_{i=1}^n \left(\frac{1}{\varepsilon} \rho_i(x), \infty \right), \end{aligned}$$

giving $\inf(J) = \frac{1}{\varepsilon} \max\{\rho_i(x) : 1 \leq i \leq n\}$. Therefore, if $\lambda \in J$, then $\tau(x) \leq \lambda$, whence $\tau(x) \leq \inf(J)$. Therefore, we have shown that $\tau(x) \leq \frac{1}{\varepsilon} \max\{\rho_i(x) : 1 \leq i \leq n\}$.

In the reverse direction, suppose $(*)$ holds. Let $x \in V$, $\rho_1, \dots, \rho_n \in \mathcal{N}$, and let $\eta > 0$. Let

$$y \in \bigcap_{i=1}^n U(x, \rho_i, \eta).$$

By the reverse triangle inequality, we have

$$\begin{aligned} |\tau(y) - \tau(x)| &\leq \tau(y - x) \\ &< \varepsilon, \end{aligned}$$

but

$$\begin{aligned} \tau(y - x) &\leq C \max\{\rho_i(y - x) : 1 \leq i \leq n\} \\ &\leq C\eta \\ &< \varepsilon \end{aligned}$$

if η is small enough. Thus, we have for all $y \in \bigcap_{j=1}^n U(x, \rho_j, \eta)$, then $|\tau(y) - \tau(x)| < \varepsilon$, whence τ is continuous at x . $\square$

Remark: If $\tau \in \mathcal{N}$, then τ is continuous with respect to $\mathcal{T}(\mathcal{N})$.

Proposition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. Then, $U \subseteq V$ is open if and only if for all $x \in U$, there is some continuous seminorm τ and some $\varepsilon > 0$ such that $U(x, \tau, \varepsilon) \subseteq U$.

Theorem (Continuity of Multilinear Maps on locally convex topological vector spaces): Suppose $n \geq 1$, and $V_1, \dots, V_n, W$ are $\mathbb{K}$ -locally convex topological vector spaces, and let $M: V_1 \times \dots \times V_n \rightarrow W$ be a $\mathbb{K}$ -multilinear map.

Then, M is (jointly) continuous if and only if for all continuous seminorm τ on W , there exist $\rho_1, \dots, \rho_n$ continuous seminorms on $V_1, \dots, V_n$ respectively such that for all tuples $(x_1, \dots, x_n) \in V_1 \times \dots \times V_n$,

$$\tau(M(x_1, \dots, x_n)) \leq \rho(x_1) \cdots \rho(x_n). \quad (\dagger)$$

Proof. Suppose M is (jointly) continuous. That is, M is equipped with the product topology. Since τ is a continuous seminorm, we have that τ is a continuous map from $W \rightarrow [0, \infty)$, we have $\tau^{-1}([0, 1))$ is open in W . Moreover, $M(0, \dots, 0) = 0 \in \tau^{-1}([0, 1))$. Since M is continuous at $(0, \dots, 0)$, there is a neighborhood $U_1 \times \dots \times U_n$ in $V_1 \times \dots \times V_n$ where each U_i is a neighborhood of 0 in V_i , such that $M(U_1 \times \dots \times U_n) \subseteq \tau^{-1}([0, 1))$.

Let $(x_1, \dots, x_n) \in V_1 \times \dots \times V_n$, and let $\lambda_1, \dots, \lambda_n > 0$. We have

$$\begin{aligned} \tau(M(x_1, \dots, x_n)) &= \tau(M(\lambda_1 \lambda_1^{-1} x_1, \dots, \lambda_n \lambda_n^{-1} x_n)) \\ &= \lambda_1 \cdots \lambda_n \tau(M(\lambda_1^{-1} x_1, \dots, \lambda_n^{-1} x_n)) \\ &< \lambda_1 \cdots \lambda_n \end{aligned}$$

if for all i , we have $\lambda_i^{-1} x_i \in U_i$. For each $i \in [n]$, we have U_i is an open neighborhood of $0 \in V_i$, so there exists some continuous seminorm σ_i and $\varepsilon_i > 0$ such that $U(0, \sigma_i, \varepsilon_i) \subseteq U_i$. If for every i , we have $\sigma_i(\lambda_i^{-1} x_i) < \varepsilon_i$, then

$$\tau(M(x_1, \dots, x_n)) < \lambda_1 \cdots \lambda_n.$$

Rewriting, we have $\lambda_i > \frac{1}{\varepsilon_i} \sigma_i(x_i)$ for each such i , so that we have

$$\begin{aligned} \tau(M(x_1, \dots, x_n)) &\leq \lambda_1 \cdots \lambda_n \\ &\leq \inf \left\{ \lambda_1 \cdots \lambda_n : \lambda_i > 0 \text{ and for all } i \in [n], \lambda_i > \frac{1}{\varepsilon_i} \sigma_i(x_i) \right\} \\ &= \left(\frac{1}{\varepsilon_1} \sigma_1(x_1) \right) \cdots \left(\frac{1}{\varepsilon_n} \sigma_n(x_n) \right) \end{aligned}$$

Setting $\rho_i = \frac{1}{\varepsilon_i} \sigma_i$ yields the desired result.

Assume that $(\dagger)$ holds. Let $(x_1, \dots, x_n), (y_1, \dots, y_n) \in V_1 \times \dots \times V_n$. We want to try to compare the difference between $M(x_1, \dots, x_n)$ and $M(y_1, \dots, y_n)$. From multilinearity, we have

$$\begin{aligned} M(x_1, \dots, x_n) - M(y_1, \dots, y_n) &= \sum_{i=1}^n (M(y_1, \dots, y_{i-1}, y_i, x_{i+1}, \dots, x_n) - M(y_1, \dots, y_{i-1}, x_i, x_{i+1}, \dots, x_n)) \\ &= \sum_{i=1}^n M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n) \end{aligned}$$

Let $Z \subseteq W$ be open containing $z = M(x_1, \dots, x_n)$, and write $z' = M(y_1, \dots, y_n)$. Then, there exists a continuous seminorm τ on W and $\varepsilon > 0$ such that $U(z, \tau, \varepsilon) \subseteq Z$.

Consider $\tau(z - z')$. Then, we have by multilinearity and the triangle inequality that

$$\tau(z - z') \leq \sum_{i=1}^n \tau(M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n)).$$

Using $(\dagger)$ gives continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that

$$\leq \sum_{i=1}^n \rho_1(y_1) \cdots \rho_{i-1}(y_{i-1}) \rho_i(y_i - x_i) \rho_{i+1}(x_{i+1}) \cdots \rho_n(x_n)$$

$$\leq n(C + \eta)^{n-1}\eta,$$

where we assume that for all i , $y_i \in U(x_i, \rho_i, \eta)$, and $C = \max\{\rho_i(x_i) : i = 1, \dots, n\}$. By choosing η sufficiently small, $n(C + \eta)^{n-1}\eta < \varepsilon$, which yields

$$M\left(\prod_{i=1}^n U(x_i, \rho_i, \eta)\right) \subseteq U(z, \tau, \varepsilon) \subseteq Z,$$

so M is continuous. $\square$

Theorem (Refinement of Previous Theorem): Let $V_1, \dots, V_n$ and W be locally convex topological vector spaces, and let $\mathcal{N}$ be a defining collection of seminorms for the topology on W . Then, if $M: V_1 \times \dots \times V_n \rightarrow W$ is a multilinear map, we have M is continuous if and only if for every $\tau \in \mathcal{N}$, there exist continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that for all $x = (x_1, \dots, x_n)$,

$$\tau(M(x)) \leq \prod_{i=1}^n \rho_i(x_i).$$

Proof. The forward direction follows from the fact that τ is a continuous seminorm, while the reverse direction follows from the fact that any continuous seminorm is dominated by a constant multiple of some family of seminorms in $\mathcal{N}$, and a constant multiple of the continuous seminorm ρ_i on V_i is also a continuous seminorm on V_i . $\square$

Corollary: If $T: V \rightarrow W$ is a linear map of locally convex topological vector spaces $\mathcal{N}$ defining the topology on W , then T is continuous if and only if for all $\tau \in \mathcal{N}$ (or all continuous seminorms on W), there exists a continuous seminorm ρ on V such that for all $x \in V$, $\tau(T(x)) \leq \rho(x)$.

Proposition: Let V be a $\mathbb{K}$ -vector space, and let $\mathcal{N}_1, \mathcal{N}_2 \subseteq \text{SN}(V)$. Then, $\mathcal{T}(\mathcal{N}_1) \subseteq \mathcal{T}(\mathcal{N}_2)$ if and only if for all $\tau \in \mathcal{N}_1$, τ is a continuous seminorm for the locally convex topological vector space $(V, \mathcal{T}(\mathcal{N}_2))$.

Proof. We have that $\mathcal{T}(\mathcal{N}_1)$ is contained in $\mathcal{T}(\mathcal{N}_2)$ if and only if the identity map $\text{id}: (V, \mathcal{T}(\mathcal{N}_2)) \rightarrow (V, \mathcal{T}(\mathcal{N}_1))$ is continuous. $\square$

Remark: We recover the fact that $(V, \|\cdot\|_1)$ and $(V, \|\cdot\|_2)$ are equivalent as norms if and only if the identity map $\text{id}: (V, \|\cdot\|_1) \rightarrow (V, \|\cdot\|_2)$ is a homeomorphism.

Normability and Metrizability

We will focus on Hausdorff locally convex topological vector spaces from now on. That is, the defining family of seminorms separates points.

Definition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a Hausdorff locally convex topological vector space. We say V is normable if and only if there exists a norm $\|\cdot\|$ on V such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\|\cdot\|)$.

Theorem: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. The following are equivalent:

- (i) $(V, \mathcal{T}(\mathcal{N}))$ is normable;
- (ii) there exists a finite subset F of $\mathcal{T}(\mathcal{N})$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(F)$;

Proof. We start by showing that (i) implies (ii). Let $\|\cdot\|$ be a norm such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\|\cdot\|)$. Since $\mathcal{T}(\|\cdot\|) \subseteq \mathcal{T}(\mathcal{N})$, there exist $\rho_1, \dots, \rho_n \in \mathcal{N}$ and some constant C such that

$$\|\cdot\| \leq C \max\{\rho_k : k = 1, \dots, n\}.$$

Let $\tau \in \mathcal{N}$. Then, since $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\|\cdot\|)$, it follows that there is some K such that $\tau \leq K\|\cdot\|$, whence

$$\tau \leq CK \max\{\rho_k : k = 1, \dots, n\}$$

In particular, this means that the finite subset of seminorms $F = \{\rho_1, \dots, \rho_n\}$ generates the topology $\mathcal{T}(\mathcal{N})$ since every seminorm in $\mathcal{N}$ is continuous with respect to $\mathcal{T}(F)$.

Next, we show that (ii) implies (i). Suppose there is a finite subset $F \subseteq \mathcal{N}$ with elements $\rho_1, \dots, \rho_n$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(F)$. In particular, we have that for all seminorms $\tau \in \mathcal{N}$, we have

$$\tau \leq C \max\{\rho_k : k = 1, \dots, n\}.$$

We claim that

$$\|\cdot\| = \max\{\rho_k : k = 1, \dots, n\}$$

induces the topology on $\mathcal{N}$. This follows from the fact that every seminorm $\tau \in \mathcal{N}$ is continuous with respect to $\|\cdot\|$, so $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\|\cdot\|) \subseteq \mathcal{T}(\mathcal{N})$, with the latter inclusion emerging from the fact that $\|\cdot\|$ is necessarily a continuous seminorm in $\mathcal{N}$ by the way it was defined. Since $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff, it follows that $\|\cdot\|$ separates points, so $\|\cdot\|$ is a norm. $\square$

This allows us to prove that certain examples we discussed earlier are not normable spaces.

Example: Consider

$$V = \mathbb{R}^{\mathbb{N}}$$

with topology defined by seminorms $\rho_n = |x_n|$. Then, we observe that there are no finite subsets of the seminorms $\mathcal{N} = \{\rho_k : k \in \mathbb{N}\}$ that separate the points of V by virtue of the fact that if $F = \{\rho_{i_1}, \dots, \rho_{i_j}\}$ is a finite subset, then the elementary vector e_{i_j+1} with 1 in position $(i_j + 1)$ is taken to 0 by any continuous seminorm on $\mathcal{T}(F)$, but e_{i_j+1} is not zero.

Next, we concern ourselves with metrizability of topological vector spaces.

Recall that (X, τ) is *first countable* if for any $x \in X$, the system of open neighborhoods $\mathcal{O}_x$ has a countable basis (i.e., a countable neighborhood basis). That is, there is some $\{U_n : n \geq 1\} \subseteq \mathcal{O}_x$ such that for any $Z \in \mathcal{O}_x$, there is some n such that $U_n \subseteq Z$. Metric spaces are always first countable by virtue of the Archimedean property.

Proposition: A (not necessarily locally convex) TVS $(V, \mathcal{T})$ is first countable if and only if 0 has a countable neighborhood basis.

Proof. Addition by a fixed vector $v \in V$ is a homeomorphism of V . $\square$

Theorem: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. The following are equivalent:

- (i) the space $(V, \mathcal{T}(\mathcal{N}))$ is metrizable;
- (ii) the space $(V, \mathcal{T}(\mathcal{N}))$ is first countable;
- (iii) there exists a countable subset $\mathcal{M} \subseteq \mathcal{N}$ such that $\mathcal{T}(\mathcal{M}) = \mathcal{T}(\mathcal{N})$.

Proof. We see that (i) implies (ii) since metrizable spaces are first countable.

Next, we show that (ii) implies (iii). Let $\{U_n : n \geq 0\}$ be a neighborhood basis for 0. Then, for any n , there exists finite $F_n \subseteq \mathcal{N}$ and $\varepsilon_n > 0$ such that

$$\bigcap_{\rho \in F_n} U(0, \rho, \varepsilon_n) \subseteq U_n.$$

Taking $\mathcal{M} = \bigcup_{n \geq 0} F_n$, this yields a countable subcollection of $\mathcal{N}$. We will show that $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\mathcal{M})$. That is, we will have to show that for all $\tau \in \mathcal{N}$, τ is continuous with respect to $\mathcal{M}$.

We have $\tau^{-1}([0, 1])$ is a neighborhood of 0 in $\mathcal{T}(\mathcal{N})$. This contains U_n for some $n \in \mathbb{N}$, and thus some finite subset F_n with

$$\begin{aligned} \bigcap_{\rho \in F_n} U(0, \rho, \varepsilon_n) &\subseteq U_n \\ &\subseteq \tau^{-1}([0, 1]). \end{aligned}$$

In particular, this means that if $x \in V$ is such that for all $\rho \in F_n$, $\rho(x) < \varepsilon_n$, then $x \in \tau^{-1}([0, 1))$, whence τ is continuous. Scaling allows one to apply the same method for any $x \in V$.

Finally, we will show that (iii) implies (i). Let $\mathcal{M} \subseteq \mathcal{N}$ be a countable subset with seminorms $\{\rho_n : n \geq 0\}$. Let

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min\{\rho_n(x - y), 1\}.$$

Then, this is a distance metric and $T(d) = T(\mathcal{M})$. □

Definition: A locally convex topological vector space (V, T) . If (V, T) is complete metrizable, then we call V a *Fréchet space*.

Example:

- (i) Banach spaces are Fréchet spaces.
- (ii) The space of rapidly decreasing sequences $s(\mathbb{N}, \mathbb{K})$ is a Fréchet space that is not normable. The distance metric is given by

$$d(x, y) = \sum_{k=0}^{\infty} 2^{-k} \min(1, \rho_k(x - y)).$$

- (iii) The vector space $V = \mathbb{R}^{\mathbb{N}}$ with the product topology is a Fréchet space.
- (iv) The space of finitely supported sequences $c_{00}(\mathbb{N}, \mathbb{K})$, henceforth denoted c_{00} , with the finest locally convex topology is a topological vector space that is not metrizable, let alone a Fréchet space. This follows from a diagonal argument on the assumption of countably many seminorms.

Consider the uncountable collection of seminorms $\mathcal{N}$ given by, for any $\omega \in [0, \infty)^{\mathbb{N}}$,

$$\rho_{\omega}(x) = \sum_{n=0}^{\infty} \omega_n |x_n|,$$

where we observe that each ρ_{ω} is distinct. It is possible to show that $T(\mathcal{N}) = T(\text{SN}(c_{00}))$.

One direction follows directly from the definition. The reverse direction uses the fact that $\{e_n : n \geq 0\}$ forms an algebraic basis for c_{00} . If ρ is a seminorm, then we will define $\omega_n = \rho(e_n)$. For any $x \in c_{00}$, then we may write

$$x = \sum_{n=0}^N x_n e_n,$$

satisfying

$$\begin{aligned} \rho(x) &= \rho\left(\sum_{n=0}^N x_n e_n\right) \\ &\leq \sum_{n=0}^N \omega_n |x_n| \\ &= \rho_{\omega}(x). \end{aligned}$$

Since x is arbitrary, it follows that every seminorm is continuous with respect to the seminorms of the family $\mathcal{N}$, whence the family $\mathcal{N}$ defines the topology on c_{00} .

In fact, we may assume that $\omega \in \mathbb{N}^{\mathbb{N}}$ by replacing $(\omega_n)_n \in [0, \infty)^{\mathbb{N}}$ with $(\lceil \omega_n \rceil)_n$.

Notable Examples

Example: Let $K \subseteq \mathbb{R}^d$ be a compact subset. Define

$$\mathcal{D}_K(\mathbb{R}^d) = C_K^\infty(\mathbb{R}^d)$$

to be the space of C^∞ with $\text{supp}(f) \subseteq K$. Define seminorms for each $N \geq 0$ by

$$\rho_N(f) = \max_{\substack{\alpha \in \mathbb{N}^d \\ |\alpha| \leq N}} \sup_{x \in \mathbb{R}^d} |\partial^\alpha f(x)|.$$

Let $\mathcal{T} = \mathcal{T}(\{\rho_N : N \geq 0\})$. This is a countable collection of seminorms, so the space $C_K^\infty(\mathbb{R}^d)$ equipped with $\mathcal{T}$ is a Fréchet space with the metric

$$d(f, g) = \sum_{N \geq 0} 2^{-N} \min(1, \rho_N(f - g)).$$

Example: A function $f: \mathbb{R}^d \rightarrow \mathbb{C}$ is called a Schwartz function if $f \in C^\infty(\mathbb{R}^d)$ and for all $\alpha \in \mathbb{N}^d$ and all $k \in \mathbb{N}$,

$$\sup_{x \in \mathbb{R}^d} \langle x \rangle^k |\partial^\alpha f(x)| < \infty.$$

We label these seminorms by $\rho_{\alpha, k}(f)$. The space $\mathcal{S}(\mathbb{R}^d)$ of all such functions is known as the *Schwartz space* for $\mathbb{R}^d$.

The standard topology on $\mathcal{S}(\mathbb{R}^d)$ is given by $\mathcal{T} = \mathcal{T}(\mathcal{N})$, where $\mathcal{N}$ is the collection of all such $\rho_{\alpha, k}$. We may also write $\|\cdot\|_{\alpha, k}$ for $\rho_{\alpha, k}$.

We can also consider the family of all such ρ_N with

$$\rho_N(f) = \max_{\substack{|\alpha| \leq N \\ k \leq N}} \rho_{\alpha, k}(f),$$

which yields the metric

$$d(f, g) = \sum_{N=0}^{\infty} 2^{-N} \min(1, \rho_N(f - g)).$$

This induces the same topology as the family of all the $\rho_{\alpha, k}$.

With this distance metric, the Schwartz space is complete. In particular, this means that $\mathcal{S}(\mathbb{R}^d)$ is an example of a Fréchet space. The Schwartz space is needed to define the Fourier transform (which we will discuss more heavily later). The Fourier transform is defined by

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-ix\xi} f(x) d^d x.$$

When $f \in L^1$, we have that $\hat{f}$ is well-defined, but $\hat{f}$ is not necessarily in L^1 . It turns out that the Fourier transform is an automorphism of the Schwartz space (over $\mathbb{C}$). Note that Schwartz functions are L^1 since they decay faster than functions of the form

$$u(x) = \langle x \rangle^\lambda$$

where $\lambda < d$. That such a $u(x)$ is integrable can be found by using the hyperspherical coordinates.

Example (A Non-Example): The dual space of the Schwartz space, $\mathcal{S}(\mathbb{R}^d)^*$, is called the space of *tempered distributions*. We say $\mathcal{S}(\mathbb{R}^d)^* \subseteq C_c^\infty(\mathbb{R}^d)^*$, where the dual space $C_c^\infty(\mathbb{R}^d)^*$ is called the space of all distributions. We will discuss these more soon.

The space $C_c^\infty(\mathbb{R}^d)$ is the space of compactly supported C^∞ functions. In particular,

$$\begin{aligned} C_c^\infty(\mathbb{R}^d) &= \bigcup_{K \text{ compact}} C_K^\infty(\mathbb{R}^d) \\ &= \bigcup_{N \geq 1} C_{B(0,N)}^\infty(\mathbb{R}^d), \end{aligned}$$

where

$$B(0, N) = \{x \in \mathbb{R}^d : |x| \leq N\}$$

is the closed ball centered at 0.

Though $C_c^\infty(\mathbb{R}^d)$ is the increasing countable union of Fréchet spaces, it is not itself Fréchet; in fact, $C_c^\infty(\mathbb{R}^d)$ is not even metrizable. It is an example of an inductive limit of a family of Fréchet spaces.

Let

$$\iota_K : C_K^\infty(\mathbb{R}^d) \rightarrow C_c^\infty(\mathbb{R}^d)$$

denote the inclusion maps. We say a seminorm ρ on $C_c^\infty(\mathbb{R}^d)$ is compatible if and only if for every compact $K \subseteq \mathbb{R}^d$, $\rho \circ \iota_K$ is a continuous seminorm on $C_K^\infty(\mathbb{R}^d)$. The topology on $C_c^\infty(\mathbb{R}^d)$ is the topology generated by all compatible seminorms.

Definition: We say a sequence $(f_n)_n$ of functions on $U \subseteq \mathbb{R}^d$ converges locally uniformly to g if for all $x \in U$, there is an open subset $V \subseteq U$ containing x such that

$$\lim_{n \rightarrow \infty} \sup_{y \in V} |f_n(y) - g(y)| = 0.$$

Theorem: The sequence $(f_n)_n$ converges locally uniformly to g if and only if $(f_n)_n$ converges uniformly on compact subsets.

Proposition: Let $(f_n)_n$ be a sequence of C^1 functions on $\mathbb{R}^d$, and suppose f_n converges pointwise to $f : U \rightarrow \mathbb{C}$. Furthermore, assume that for all $i \in [d]$, $\partial_i f_n$ converges locally uniformly to some function $g_i : U \rightarrow \mathbb{C}$. Then, f is C^1 and $g_i = \partial_i f$ for all i .

Proof. Since each g_i is the local uniform limit of some continuous functions, each g_i is continuous. Fix $x \in U$.

Since $\partial_i f_n \rightarrow g_i$ locally uniformly, there is some $\varepsilon > 0$ such that

$$\sup_{y \in U(x, \varepsilon)} |\partial_i f_n(y) - g_i(y)| \xrightarrow{n \rightarrow \infty} 0.$$

For some $t \in \mathbb{R}$ with $0 < |t| < \varepsilon$, the fundamental theorem of calculus gives

$$f_n(x + te_i) - f_n(x) = t \int_0^1 \partial_i f_n(x + ste_i) ds.$$

Taking the limit as $n \rightarrow \infty$ and using dominated convergence gives

$$f(x + te_i) - f(x) = \lim_{n \rightarrow \infty} t \int_0^1 \partial_i f_n(x + ste_i) ds$$

$$= t \int_0^1 g_i(x + ste_i) ds.$$

Therefore, we have

$$\begin{aligned} \left| \frac{f(x + te_i) - f(x)}{t} - g_i(x) \right| &= \left| \int_0^1 g_i(x + ste_i) - g_i(x) ds \right| \\ &\leq \sup_{0 \leq s \leq 1} |g_i(x + ste_i) - g_i(x)| \\ &\rightarrow 0 \end{aligned}$$

by the continuity of g_i at x . Thus, $\partial_i f = g_i$, so f is C^1 . $\square$

Duality

Let $(V, \mathcal{T})$ be a topological vector space. The algebraic dual space V' is the space of all linear functionals $L: V \rightarrow \mathbb{C}$. The topological dual space, V^* , is the space of all continuous linear functionals $L: V \rightarrow \mathbb{C}$.

Definition: The weak* topology on V^* is the topology induced by the family of seminorms

$$\mathcal{N}_{\text{wk}^*} := \{(L \mapsto |L(x)|) : x \in V\}.$$

There is another locally convex topology on V^* that is the analogue of the norm topology in the case of a normed vector space. To understand this analogue, we must discuss boundedness.

Definition: Let $(V, \mathcal{T})$ be a topological vector space. We say A is bounded if for all open neighborhoods $U \in \mathcal{O}_0$ of 0, there is $\lambda > 0$ such that $\lambda A \subseteq U$.

Theorem: Let V be a normed vector space. Then, A is bounded in the above definition if and only if A is bounded in the usual sense.

Proposition: Let $(V, \mathcal{T})$ be a topological vector space, and let $A \subseteq V$. The following are equivalent:

- (i) A is bounded;
- (ii) for all $\rho \in \mathcal{N}$, $\sup_{x \in A} \rho(x) < \infty$;
- (iii) for all continuous seminorms ρ , $\sup_{x \in A} \rho(x) < \infty$.

Proof. Assume (i); that is, suppose A is bounded, and let ρ be a defining seminorm. Then, there exists some $\lambda > 0$ such that $\lambda A \subseteq U(0, \rho, 1)$. That is, for all $x \in A$,

$$\lambda \rho(x) < 1,$$

or that $\rho(x) < \lambda^{-1}$. Therefore, $\sup_{x \in A} \rho(x) \leq \lambda^{-1}$, giving (ii).

Suppose (ii) holds. Let ρ be a continuous seminorm on $(V, \mathcal{T})$. Then, we may write $\rho(x) \leq C \max\{\rho_i(x) : i = 1, \dots, n\}$ for some family of defining seminorms ρ_i . This gives (iii).

Assume (iii) holds. Let $U \in \mathcal{O}_0$ be an open neighborhood of 0. Then, there exists some continuous seminorm ρ and some $\varepsilon > 0$ such that $U(0, \rho, \varepsilon) \subseteq U$. Notice that for any $x \in A$ and any $\lambda > 0$, we have

$$\begin{aligned} \rho(\lambda x) &= \lambda \rho(x) \\ &\leq \lambda \sup_{y \in A} \rho(y). \end{aligned}$$

Since $M = \sup_{y \in A} \rho(y)$ is finite by assumption, choosing λ sufficiently small such that $0 < \lambda < \varepsilon/M$ yields $\rho(\lambda x) < \varepsilon$ for all $x \in A$, or that $\lambda A \subseteq U(0, \rho, \varepsilon)$. $\square$

Theorem: Let $(V, \mathcal{T})$ be a topological vector space. Then, $\mathcal{T}$ is a norm topology if and only if V contains a bounded nonempty open neighborhood of 0.

Definition: Let $(V, \mathcal{T})$ be a topological vector space. The *strong dual* is the topology on V^* defined by the family of all ρ_A with A bounded, where

$$\rho_A(L) = \sup_{x \in A} |L(x)|.$$

Definition: Let V and W be topological vector spaces with strong duals V^* and W^* respectively. Let $\varphi: V \rightarrow W$ be a continuous linear map. Then, the transpose is given by $\varphi^*: W^* \rightarrow V^*$ taking $\varphi^*(L) = L \circ \varphi$.

Recall that the dual space $C_c^\infty(\mathbb{R}^d)^*$ of compactly supported C^∞ functions is called the space of *distributions*, while the dual space $\mathcal{S}(\mathbb{R}^d)^*$ is called the space of *tempered distributions*.

Given $\theta \in \mathcal{S}(\mathbb{R}^d)^*$ and $f \in \mathcal{S}(\mathbb{R}^d)$, we write $\langle f, \theta \rangle = \theta(f)$.

Since $C_c^\infty(\mathbb{R}^d) \subseteq \mathcal{S}(\mathbb{R}^d)$, we will define a canonical inclusion map $\varphi: C_c^\infty(\mathbb{R}^d) \rightarrow \mathcal{S}(\mathbb{R}^d)$.

Theorem: The inclusion φ is continuous with respect to the locally convex topology on each space.

Fourier Analysis

The next section of our coverage is focused on Fourier analysis.

Classical Fourier Transform

We will assume that $\mathbb{K} = \mathbb{C}$.

Definition: Let

$$\begin{aligned} \mathcal{F}: \mathcal{S}(\mathbb{R}^d, \mathbb{C}) &\rightarrow \mathcal{S}(\mathbb{R}^d, \mathbb{C}) \\ f &\mapsto \hat{f} = \mathcal{F}[f] \end{aligned}$$

be defined by

$$\hat{f}(\xi) := \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} f(x) d^d x.$$

Remark: We will let

$$\mathcal{G}: \mathcal{S}(\mathbb{R}^d, \mathbb{C}) \rightarrow \mathcal{S}(\mathbb{R}^d, \mathbb{C})$$

be defined by

$$\mathcal{G}[\hat{f}](x) = \int_{\mathbb{R}^d} e^{i\langle \xi, x \rangle} \hat{f}(\xi) d^d \xi.$$

Theorem: The map $\mathcal{F}$ is well-defined, linear, and continuous.

Proof. We recall the seminorms

$$\|f\|_{\alpha, k} = \sup_{x \in \mathbb{R}^d} \langle x \rangle^k |\partial^\alpha f(x)|.$$

We want to verify that $\mathcal{F}$ is well-defined. We note that

$$\begin{aligned} \int_{\mathbb{R}^d} |e^{-i\langle x, \xi \rangle} f(x)| d^d x &= \int_{\mathbb{R}^d} |\langle x \rangle^{-(d+1)} \langle x \rangle^{d+1} f(x)| d^d x \\ &\leq \|f\|_{0, d+1} \int_{\mathbb{R}^d} \langle x \rangle^{-(d+1)} d^d x \\ &< \infty. \end{aligned}$$

Our next job is to show that $\hat{f}$ is C^∞ . By dominated convergence, we have that

$$\begin{aligned}\partial_j \hat{f}(\xi) &= \int_{\mathbb{R}^d} \frac{\partial}{\partial \xi_j} \left(e^{-i\langle x, \xi \rangle} f(x) \right) d^d x \\ &= \int_{\mathbb{R}^d} (-ix_j) e^{-i\langle x, \xi \rangle} f(x) d^d x.\end{aligned}$$

Dominated convergence applies since applying the derivative to the integrand yields the upper bound given by

$$\begin{aligned}\left| (-ix_j) e^{-i\langle x, \xi \rangle} f(x) \right| &\leq \sqrt{1 + x_1^2 + \dots + x_n^2} |f(x)| \\ &= \langle x \rangle |f(x)| \\ &=: g(x),\end{aligned}$$

which is integrable since f is a Schwartz function.

Inductively, we have

$$\left| \frac{\partial^n}{\partial \xi_{j_1} \dots \partial \xi_{j_n}} e^{-i\langle x, \xi \rangle} f(x) \right| \leq \langle x \rangle^n |f(x)|.$$

In general, for any multi-index $\alpha \in \mathbb{N}^d$, we have

$$\partial^\alpha \hat{f}(\xi) = \int_{\mathbb{R}^d} (-i)^{|\alpha|} x^\alpha e^{-i\langle x, \xi \rangle} f(x) d^d x.$$

Therefore, $\hat{f}$ is C^∞ .

Our next goal is to extract the decay rate so that $\hat{f}$ is in Schwartz space. Let $\alpha, \beta \in \mathbb{N}^d$, and consider

$$\begin{aligned}\xi^\beta \partial_\xi^\alpha \hat{f}(\xi) &= \int_{\mathbb{R}^d} (-i)^{|\alpha|} x^\alpha \xi^\beta e^{-i\langle x, \xi \rangle} f(x) d^d x \\ &= i^{|\beta| - |\alpha|} \int_{\mathbb{R}^d} \partial_x^\beta \left(e^{-i\langle x, \xi \rangle} \right) x^\alpha f(x) d^d x,\end{aligned}$$

where we note that $\partial_x^\beta \left(e^{-i\langle x, \xi \rangle} \right) = i^{-|\beta|} \xi^\beta e^{-i\langle x, \xi \rangle}$.

Integrating by parts and the Leibniz rule yield

$$\begin{aligned}\xi^\beta \partial_\xi^\alpha \hat{f}(\xi) &= (-i)^{|\alpha| + |\beta|} \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} \partial_x^\beta (x^\alpha f(x)) d^d x \\ &= (-i)^{|\alpha| + |\beta|} \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} \sum_{\substack{0 \leq \gamma \leq \beta \\ \gamma \in \mathbb{N}^d}} \frac{\beta!}{\gamma! (\beta - \gamma)!} \partial_x^\gamma (x^\alpha) \partial_x^{\beta - \gamma} (f(x)) d^d x \\ &= (-i)^{|\alpha| + |\beta|} \sum_{\gamma \in \mathbb{N}^d} \chi_{\{\gamma \leq \min(\alpha, \beta)\}} \frac{\alpha! \beta!}{\gamma! (\alpha - \gamma)! (\beta - \gamma)!} \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} d^d x.\end{aligned}$$

Estimating yields

$$\left| \xi^\beta \partial_\xi^\alpha \hat{f}(\xi) \right| \leq \sum_{\gamma \in \mathbb{N}^d} \chi_{\{\gamma \leq \min(\alpha, \beta)\}} \frac{\alpha! \beta!}{\gamma! (\alpha - \gamma)! (\beta - \gamma)!} \left(\int_{\mathbb{R}^d} \langle x \rangle^{-(d+1)} d^d x \right) \|f\|_{\beta - \gamma, |\alpha| - |\gamma| + d + 1}.$$

In particular, the right-hand side is a finite constant, so that

$$\sup_{\xi \in \mathbb{R}^d} |\langle \xi \rangle^{2m} \partial_\xi^\alpha \hat{f}(\xi)| = \sup_{\xi \in \mathbb{R}^d} \left(1 + \xi_1^2 + \dots + \xi_d^2 \right)^m |\partial_\xi^\alpha \hat{f}(\xi)|$$

$< \infty$.

Thus, $\hat{f} \in \mathcal{S}(\mathbb{R}^d)$. Furthermore, this estimate yields that $\mathcal{F}$ is continuous, as any defining seminorm applied to $\hat{f}$ is bounded by a constant multiple of a seminorm applied to f .

Similarly, the map $\mathcal{G}$ simply substitutes i for $-i$, so that $\mathcal{G}$ is also continuous. $\square$

The most important Fourier transform that we will compute is the Gaussian. That is, $f(x) = e^{-x^2/2}$.

Theorem: If $f(x) = e^{-|x|^2/2}$ over $\mathbb{R}^d$, we have $\hat{f}(\xi) = (2\pi)^{d/2} e^{-\xi^2/2}$.

Proof. Notice that $f(x) \in \mathcal{S}(\mathbb{R}^d)$. Fubini's theorem yields

$$\begin{aligned} \hat{f}(\xi) &= \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} e^{-|x|^2/2} d^d x \\ &= \int_{\mathbb{R}^d} \prod_{j=1}^d e^{-i\xi_j x_j - (x_j^2/2)} d^d x \\ &= \prod_{j=1}^d \int_{\mathbb{R}} e^{-x^2/2 - i\xi x} dx. \end{aligned}$$

Therefore, we reduce to the one-dimensional case. One can complete the square, yielding

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-x^2/2 - i\xi x} dx &= e^{-\xi^2/2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x+i\xi)^2} dx \\ &= e^{-\xi^2/2} \lim_{A \rightarrow \infty} \int_{-A}^A e^{-\frac{1}{2}(x+i\xi)^2} dx \end{aligned}$$

Translating by $-i\xi$ and using the residue theorem on the resulting rectangular contour yields that the integral is the same as the original Gaussian integral.

The second method follows by a Taylor expansion. By dominated convergence, we have

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-x^2/2 - i\xi x} dx &= \int_{-\infty}^{\infty} x^n e^{-x^2/2} \sum_{n=0}^{\infty} \frac{(i\xi)^n}{n!} dx \\ &= \sum_{n=0}^{\infty} \frac{(i\xi)^n}{n!} \int_{-\infty}^{\infty} x^n e^{-x^2/2} dx. \end{aligned}$$

If X is a random variable with standard normal distribution, we note that the integral evaluates to

$$\int_{-\infty}^{\infty} x^n e^{-x^2/2} dx = (2\pi)^{1/2} \mathbb{E}[X^n],$$

while

$$\mathbb{E}[X^n] = \begin{cases} 0 & n \text{ is odd} \\ \frac{(2n)!}{2^n n!} & n \text{ is even.} \end{cases}$$

In particular, we have that

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-x^2/2 - i\xi x} dx &= (2\pi)^{1/2} \sum_{k=0}^{\infty} \frac{(-i)^{2k} \xi^{2k}}{(2k)!} \frac{(2k)!}{2^k k!} \\ &= (2\pi)^{1/2} e^{-\xi^2/2}. \end{aligned}$$

$\square$

Our goal is to prove the Fourier Inversion Theorem.

Theorem (Fourier Inversion Theorem): Let $\mathcal{F}$ and $\mathcal{G}$ be defined by

$$\begin{aligned}\mathcal{F}[f](\xi) &= \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} f(x) d^d x \\ \mathcal{G}[\hat{f}](x) &= \int_{\mathbb{R}^d} e^{i\langle x, \xi \rangle} \hat{f}(\xi) d^d \xi.\end{aligned}$$

Then, $\mathcal{G} \circ \mathcal{F} = \mathcal{F} \circ \mathcal{G} = (2\pi)^d$.

In particular, we obtain the following corollary.

Corollary: The map $\mathcal{F}$ is a topological vector space isomorphism from $\mathcal{S}(\mathbb{R}^d, \mathbb{C})$ to itself, with inverse given by

$$\mathcal{F}^{-1}[f](x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i\langle x, \xi \rangle} f(\xi) d^d \xi.$$

We will write $\check{f}(x)$ for $\mathcal{F}^{-1}[f](x)$.

To prove the Fourier Inversion Theorem, we will need the following proposition.

Proposition: We have

$$\lim_{\varepsilon \searrow 0} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} = \delta_x^d(y)$$

in $\mathcal{S}(\mathbb{R}^d, \mathbb{C})^*$ with the strong topology. Here, δ_x^d is the evaluation map implemented by the duality pairing

$$\langle f, \delta_x^d \rangle = \int_{\mathbb{R}^d} f(y) \delta_x^d(y) d^d y.$$

Remark: The duality pairing for $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{C})$ and $\varphi \in \mathcal{S}(\mathbb{R}^d, \mathbb{C})^*$ written as

$$\langle f, \varphi \rangle = \int f(y) \varphi(y) d^d y$$

is not really an integral. However, physicists like writing it this way.

Proof of Proposition. Let $\varphi_{x,\varepsilon} \in \mathcal{S}(\mathbb{R}^d, \mathbb{C})$ be defined by

$$\varphi_{x,\varepsilon}(y) = (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}.$$

There is a canonical injection $\iota: \mathcal{S}(\mathbb{R}^d, \mathbb{C}) \rightarrow \mathcal{S}(\mathbb{R}^d, \mathbb{C})^*$.

Let $I = [0, \infty)$ be directed by taking $\varepsilon_1 \leq \varepsilon_2$ if and only if $\varepsilon_2 \leq \varepsilon_1$. Then, our goal is to show that the net $(\iota(\varphi_{x,\varepsilon}))_{\varepsilon \in I}$ converges to δ_x^d in the strong topology.

We want to show that, for all bounded subsets $A \subseteq \mathcal{S}(\mathbb{R}^d, \mathbb{C})$, one has

$$\begin{aligned}0 &= \lim_{\varepsilon \searrow 0} \|\iota_{\varphi_{x,\varepsilon}} - \delta_x^d\|_A \\ &= \sup_{f \in A} |\langle f, \iota(\varphi_{x,\varepsilon}) - \delta_x^d \rangle|.\end{aligned}$$

We see that

$$\langle f, \iota(\varphi_{x,\varepsilon}) - \delta_x^d \rangle = \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y) f(y) d^d y - f(x)$$

$$= \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y)(f(y) - f(x)) d^d y,$$

where we note that $\varphi_{x,\varepsilon}$ was specifically chosen to have integral 1 over $\mathbb{R}^d$. Using the Fundamental Theorem of Calculus, we have

$$\begin{aligned} &= \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y) \int_0^1 \langle y - x, \nabla f(x + t(y - x)) \rangle dt d^d y \\ &= \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \int_0^1 \langle (y - x), \nabla f(x + t(y - x)) \rangle dt d^d y \\ &= \sum_{j=1}^d \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \int_0^1 \langle y_j - x_j, \partial_j f(x + t(y - x)) \rangle dt d^d y. \end{aligned}$$

Estimating the inner integral yields

$$\begin{aligned} \left| \langle f, \iota(\varphi_{x,\varepsilon}) - \partial_x^d \rangle \right| &\leq \sum_{j=1}^d \|f\|_{e_j,0} \int_{\mathbb{R}^d} |y_j - x_j| (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} d^d y \\ &\leq d \max_{1 \leq j \leq d} \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} |y - x| d^d y. \end{aligned}$$

Substituting $u = \frac{y-x}{\sqrt{\varepsilon}}$ yields $y = x + \sqrt{\varepsilon}u$, so that

$$\leq d \left(\max_{1 \leq j \leq d} \sup_{f \in A} \|f\|_{e_j,0} \right) \sqrt{\varepsilon} \left(\int_{\mathbb{R}^d} (2\pi)^{-d/2} e^{-\frac{|u|^2}{2}} |u| d^d u \right).$$

Since the integral is finite, we obtain that $\|\iota(\varphi_{x,\varepsilon}) - \delta_x^d\|_A \leq C\sqrt{\varepsilon}$. $\square$

Remark: The functions of the form $(2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}$ are a special class of functions known as [mollifiers](#).

We will now prove the Fourier Inversion Theorem.

Proof of the Fourier Inversion Theorem. Let $f \in \mathcal{S}(\mathbb{R}^d, \mathbb{C})$. We compute

$$\mathcal{G} \circ \mathcal{F}[f](x) = \int_{\mathbb{R}^d} e^{i\langle x, \xi \rangle} \hat{f}(\xi) d^d \xi.$$

Since $\hat{f}$ is a Schwartz function, dominated convergence gives

$$\begin{aligned} &= \lim_{\varepsilon \searrow 0} \int e^{i\langle x, \xi \rangle - \frac{\varepsilon}{2} |\xi|^2} \hat{f}(\xi) d^d \xi \\ &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle - \frac{\varepsilon}{2} |\xi|^2} e^{-i\langle y, \xi \rangle} f(y) d^d y \right) d^d \xi \end{aligned}$$

Making use of the factor of $e^{-\frac{\varepsilon}{2} |\xi|^2}$ allows us to use Fubini's theorem to obtain

$$= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} e^{i\langle x, \xi \rangle - \frac{\varepsilon}{2} |\xi|^2} e^{-i\langle y, \xi \rangle} d^d \xi \right) f(y) d^d y.$$

Evaluating the inner integral, we substitute $\eta = \sqrt{\varepsilon}\xi$ to obtain

$$\int_{\mathbb{R}^d} e^{-i\langle y-x, \xi \rangle} e^{-\frac{\varepsilon}{2} |\xi|^2} d^d \xi = \varepsilon^{-d/2} \int_{\mathbb{R}^d} e^{-i\left\langle \frac{y-x}{\sqrt{\varepsilon}}, \eta \right\rangle} e^{-\frac{|\eta|^2}{2}} d^d \eta$$

$$\begin{aligned}
&= \varepsilon^{-d/2} \mathcal{F} \left[\eta \mapsto e^{-\frac{|\eta|^2}{2\varepsilon}} \right] \left(\frac{y-x}{\sqrt{\varepsilon}} \right) \\
&= (\varepsilon)^{-d/2} (2\pi)^{d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \\
&= (2\pi)^d \left((2\pi)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \right).
\end{aligned}$$

Rewriting as $\varphi_{x,\varepsilon}$, we have

$$\begin{aligned}
\mathcal{G} \circ \mathcal{F}[f](x) &= (2\pi)^d \lim_{\varepsilon \searrow 0} \int_{\mathbb{R}^d} f(y) \varphi_{x,\varepsilon}(y) d^d y \\
&= (2\pi)^d \delta_x^d(f) \\
&= (2\pi)^d f(x).
\end{aligned}$$

□

Theorem (Plancherel's Theorem): For any $f, g \in \mathcal{S}(\mathbb{R}^d, \mathbb{C})$, one has

$$\int_{\mathbb{R}^d} \overline{\hat{f}(\xi)} \hat{g}(\xi) d^d \xi = (2\pi)^d \int_{\mathbb{R}^d} \overline{f(x)} g(x) d^d x.$$

Proof. The left-hand side can be written as follows

$$\int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} f(x) d^d x \right) \overline{\hat{g}(\xi)} d^d \xi = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{i\langle x, \xi \rangle} \overline{f(x)} \hat{g}(\xi) d^d \xi d^d x \quad (*)$$

This follows from the estimate with Tonelli's Theorem

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |e^{i\langle x, \xi \rangle} \overline{f(x)} \hat{g}(\xi)| d^d x d^d \xi = \left(\int_{\mathbb{R}^d} |f(x)| d^d x \right) \left(\int_{\mathbb{R}^d} |\hat{g}(\xi)| d^d \xi \right)$$

and using the fact that functions on the Schwartz space are integrable. Yet again using Fubini's theorem on (*), we have

$$\begin{aligned}
\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{i\langle x, \xi \rangle} \overline{f(x)} \hat{g}(\xi) d^d \xi d^d x &= \int_{\mathbb{R}^d} \overline{f(x)} \left(\int_{\mathbb{R}^d} e^{i\langle x, \xi \rangle} \hat{g}(\xi) d^d \xi \right) d^d x \\
&= \int_{\mathbb{R}^d} \overline{f(x)} \mathcal{G}(\mathcal{F}[g])(x) d^d x \\
&= (2\pi)^d \int_{\mathbb{R}^d} \overline{f(x)} g(x) d^d x.
\end{aligned}$$

□

We note that one can show that $\mathcal{U} = (2\pi)^{-d/2} \mathcal{F}$ is an isometry on the Schwartz space $\mathcal{S}(\mathbb{R}^d, \mathbb{C})$. Since $\mathcal{S}(\mathbb{R}^d, \mathbb{C})$ is dense in $L^2(\mathbb{R}^d)$, continuity gives that there is a unique extension $\tilde{\mathcal{U}}: L^2(\mathbb{R}^d) \rightarrow L^2(\mathbb{R}^d)$. The map $\tilde{\mathcal{U}}$ is the definition of the Fourier transform on $L^2(\mathbb{R}^d)$.

Convolution and Properties of the Fourier Transform on $\mathcal{S}(\mathbb{R}^d, \mathbb{C})$

Definition: We define the space $\mathcal{O}_M(\mathbb{R}^d, \mathbb{K})$ to consist of all functions $f: \mathbb{R}^d \rightarrow \mathbb{K}$ such that f is C^∞ and for every $\alpha \in \mathbb{N}^d$, there exists $k \in \mathbb{N}$ and a constant $C \geq 0$ such that for all $x \in \mathbb{R}^d$, one has

$$|\partial^\alpha f(x)| \leq C \langle x \rangle^k.$$

These are known as the smooth functions of tempered growth.

Proposition: Let $g \in \mathcal{O}_M$. Then, for all $f \in \mathcal{S}(\mathbb{R}^d)$, $fg \in \mathcal{S}(\mathbb{R}^d)$, and one has that the map $f \mapsto fg$ is continuous.

Proof. We will assume that $|\partial^\alpha g(x)| \leq C\langle x \rangle^{k_\alpha}$.

Now, we have

$$\partial^\alpha(fg) = \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} \partial^{\alpha-\beta} f \partial^\beta g.$$

So,

$$\begin{aligned} \langle x \rangle^k |\partial^\alpha(fg)(x)| &\leq \langle x \rangle \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} |\partial^{\alpha-\beta} f(x)| C_\beta \langle x \rangle^{k_\beta} \\ &\leq \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} C_\beta \|f\|_{\alpha-\beta, k+k_\beta} \\ &< \infty. \end{aligned}$$

So, one has that $\|fg\|_{\alpha, k} < \infty$ for all $\alpha \in \mathbb{N}^d$ and all $k \in \mathbb{N}$. The seminorm criterion also gives continuity of the map $f \mapsto fg$. $\square$

Remark: There is a converse to this theorem. Namely, if $g: \mathbb{R}^d \rightarrow \mathbb{K}$ is a function such that $fg \in \mathcal{S}(\mathbb{R}^d)$ for all $f \in \mathcal{S}(\mathbb{R}^d)$.

Bumpfunctionology (for lack of a better term) gives smoothness of such a g , but showing the polynomial bound requires a bit more work.

Example: Monomials $x \mapsto x^\alpha$ satisfy

$$\partial^\beta x^\alpha = \chi_{\{\beta \leq \alpha\}} \frac{\alpha!}{(\alpha-\beta)!} x^{\alpha-\beta},$$

so since $|x_j| \leq |x| \leq \langle x \rangle$, one has

$$|\partial^\beta x^\alpha| \leq C_\beta \langle x \rangle^{k_\beta},$$

where $k_\beta = \max(0, |\alpha - \beta|)$, and $C_\beta = \chi_{\{\beta \leq \alpha\}} \frac{\alpha!}{(\alpha-\beta)!}$.

We use

$$\begin{aligned} m_\alpha: \mathcal{S}(\mathbb{R}^d, \mathbb{K}) &\rightarrow \mathcal{S}(\mathbb{R}^d, \mathbb{K}) \\ f &\mapsto x^\alpha f. \end{aligned}$$

Proposition: The operator $\partial^\alpha: \mathcal{S}(\mathbb{R}^d, \mathbb{K}) \rightarrow \mathcal{S}(\mathbb{R}^d, \mathbb{K})$ taking $f \mapsto \partial^\alpha f$ is linear and continuous. $\square$

Proof. To be written. $\square$

Theorem (Intertwining Property of Fourier Transform): As continuous linear operators on $\mathcal{S}(\mathbb{R}^d, \mathbb{C})$, one has

$$\begin{aligned} \partial^\alpha \mathcal{F} &= (-i)^{|\alpha|} \mathcal{F} \circ m_\alpha \\ m_\alpha \circ \mathcal{F} &= (-i)^{|\alpha|} \mathcal{F} \circ \partial^\alpha \end{aligned}$$

Proof. Using dominated convergence, we may bring the derivative inside the integral, so that

$$\begin{aligned}
 \partial_\xi^\alpha \hat{f}(\xi) &= \partial_\xi^\alpha \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} f(x) d^d x \\
 &= \int_{\mathbb{R}^d} \partial_\xi^\alpha (e^{-i\langle x, \xi \rangle}) f(x) d^d x \\
 &= \int_{\mathbb{R}^d} (-ix)^{|\alpha|} e^{-i\langle x, \xi \rangle} f(x) d^d x \\
 &= (-i)^{|\alpha|} \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} x^\alpha f(x) d^d x \\
 &= (-i)^{|\alpha|} (\mathcal{F} \circ m_\alpha[f])(x).
 \end{aligned}$$

For the other direction, integration by parts yields

$$\begin{aligned}
 \xi^\alpha \hat{f}(\xi) &= \int_{\mathbb{R}^d} \xi^\alpha e^{-i\langle x, \xi \rangle} f(x) d^d x \\
 &= \int_{\mathbb{R}^d} i^{|\alpha|} \partial_x^\alpha (e^{-i\langle x, \xi \rangle}) f(x) d^d x \\
 &= (-i)^{|\alpha|} \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} \partial_x^\alpha (f(x)) d^d x \\
 &= (-i)^{|\alpha|} (\mathcal{F} \circ \partial^\alpha[f])(x)
 \end{aligned}$$

□

Proposition: Pointwise multiplication $\mathcal{S}(\mathbb{R}^d, \mathbb{C}) \times \mathcal{S}(\mathbb{R}^d, \mathbb{C}) \rightarrow \mathcal{S}(\mathbb{R}^d, \mathbb{C})$ is bilinear and (jointly) continuous.

Proof. We have

$$\partial^\alpha (fg) = \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha - \beta)!} \partial^{\alpha - \beta} f \partial^\beta g$$

so

$$\begin{aligned}
 \langle x \rangle^k |\partial^\alpha (fg)(x)| &\leq \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha - \beta)!} \|f\|_{\alpha - \beta, k} \|g\|_{\beta, 0} \\
 &\leq \tau_1(f) \tau_2(g)
 \end{aligned}$$

where

$$\begin{aligned}
 \tau_1(f) &= \left(\sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \max_{\beta \leq \alpha} \frac{\alpha!}{\beta!(\alpha - \beta)!} \right) \max_{\beta \leq \alpha} \|f\|_{\beta, k} \\
 \tau_2(g) &= \max_{\beta \leq \alpha} \|g\|_{\beta, 0}.
 \end{aligned}$$

Therefore, by the seminorm criterion for continuity of multilinear maps, it follows that $(f, g) \mapsto fg$ is continuous. □

Definition: If $f, g \in \mathcal{S}(\mathbb{R}^d, \mathbb{K})$, then

$$(f * g)(x) = \int_{\mathbb{R}^d} f(y) g(x - y) d^d y$$

is called the convolution of two functions.

Note that

$$\int_{\mathbb{R}^d} |f(y)g(x-y)| d^d y \leq \int_{\mathbb{R}^d} \langle y \rangle^{-(d+1)} \|f\|_{0,d+1} \|g\|_{0,0} d^d y,$$

so this is well-defined.

Proposition: The convolution $f * g$ is C^∞ , and for every α , $\partial^\alpha(f * g) = f * (\partial^\alpha g)$.

Proof. Dominated convergence (more work here) gives

$$\begin{aligned} \partial^\alpha(f * g)(x) &= \partial_x^\alpha \int_{\mathbb{R}^d} f(y)g(x-y) d^d y \\ &= \int_{\mathbb{R}^d} f(y) \partial_x^\alpha(g(x-y)) d^d y \\ &= \int_{\mathbb{R}^d} f(y) (\partial_x^\alpha g)(x-y) d^d y \\ &= f * (\partial^\alpha g). \end{aligned}$$

□

Lemma: For all $u, v \in \mathbb{R}^d$, one has

Proof. We have

$$\begin{aligned} \langle u+v \rangle^2 &= 1 + |u+v|^2 \\ &\leq 1 + |u|^2 + |v|^2 + 2|u||v| \\ &\leq 1 + 2|u|^2 + 2|v|^2 \\ &\leq 2 + 2|u|^2 + 2|v|^2 \\ &\leq 2(1 + |u|^2)(1 + |v|^2). \end{aligned}$$

□

Theorem: The convolution $\mathcal{S}(\mathbb{R}^d, \mathbb{K}) \times \mathcal{S}(\mathbb{R}^d, \mathbb{K}) \rightarrow \mathcal{S}(\mathbb{R}^d, \mathbb{K})$ is well-defined, bilinear, and continuous.

Proof. We have that $\partial^\alpha(f * g)$ is C^∞ . Taking brackets, we get

$$\begin{aligned} \langle x \rangle^k |\partial^\alpha(f * g)| &= \langle x \rangle^k \left| \int_{\mathbb{R}^d} f(y) (\partial_x^\alpha g)(x-y) d^d y \right| \\ &\leq \langle x \rangle \int \langle y \rangle^\ell \|f\|_{0,\ell} \langle x-y \rangle^m \|g\|_{\alpha,m} d^d y, \end{aligned}$$

where ℓ and m are to be determined. Using the fact that $\langle x \rangle \leq \sqrt{2} \langle y \rangle \langle x-y \rangle$, we get

$$\langle x \rangle \int \langle y \rangle^\ell \|f\|_{0,\ell} \langle x-y \rangle^m \|g\|_{\alpha,m} d^d y \leq 2^{k/2} \|f\|_{0,\ell} \|g\|_{\alpha,m} \int_{\mathbb{R}^d} \langle y \rangle^{k-\ell} \langle x-y \rangle^{k-m} d^d y.$$

Therefore, if we take $m = k$ and $\ell = d + k + 1$, we obtain that the right hand side is a constant, so that the convolution is in Schwartz space. Furthermore, we have shown that $\|f * g\|_{\alpha,k} \leq K \|f\|_{0,\ell} \|g\|_{\alpha,m}$ for some constant $K \geq 0$, so that the map is bilinear and continuous. □

Proposition: Convolution on $\mathcal{S}(\mathbb{R}^d, \mathbb{K})$ satisfies

- (i) $f * g = g * f$;
- (ii) $\partial(f * g) = f * (\partial g)$;
- (iii) $\partial^\alpha(f * g) = (\partial^\alpha f) * g$.

Proof.

(i) Taking a substitution of $z = x - y$ has Jacobian equal to 1, so that

$$\begin{aligned} f * g(x) &= \int_{\mathbb{R}^d} f(y)g(x-y) d^d y \\ &= \int_{\mathbb{R}^d} f(x-z)g(z) d^d z \\ &= g * f. \end{aligned}$$

(ii) Shown earlier.

(iii) Commutativity.

□

Theorem: For all $f, g \in \mathcal{S}(\mathbb{R}^d, \mathbb{C})$, one has

$$\mathcal{F}[f * g] = \mathcal{F}[f]\mathcal{F}[g].$$

Proof. Using Fubini's Theorem, we have

$$\mathcal{F}[f * g](\xi) = \int_{\mathbb{R}^d} e^{-i\langle x, \xi \rangle} \int_{\mathbb{R}^d} f(y)g(x-y) d^d y d^d x.$$

The proper estimate is found by using Tonelli's theorem so that

$$\begin{aligned} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(y)g(x-y)| d^d y d^d x &= \int_{\mathbb{R}^d} |f(y)| \int_{\mathbb{R}^d} |g(x-y)| d^d x d^d y \\ &= \left(\int_{\mathbb{R}^d} |f(y)| d^d y \right) \left(\int_{\mathbb{R}^d} |g(z)| d^d z \right) \\ &< \infty. \end{aligned}$$

Therefore, Fubini's Theorem and the substitution $z = x - y$ gives

$$\begin{aligned} \mathcal{F}[f * g](\xi) &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{-i\langle \xi, y \rangle} e^{-i\langle \xi, x-y \rangle} f(y)g(x-y) d^d x d^d y \\ &= \left(\int_{\mathbb{R}^d} e^{-i\langle \xi, y \rangle} f(y) d^d y \right) \left(\int_{\mathbb{R}^d} e^{-i\langle \xi, z \rangle} g(z) d^d z \right) \\ &= \mathcal{F}[f]\mathcal{F}[g]. \end{aligned}$$

□

Operations and Fourier Transform for $\mathcal{S}(\mathbb{R}^d, \mathbb{K})^*$

First, we define the derivative of a distribution $\varphi \in \mathcal{S}(\mathbb{R}^d, \mathbb{K})^*$, we want to define $\partial^\alpha \varphi$.

First, we work like physicists. We will write

$$\begin{aligned} \varphi(f) &= \langle f, \varphi \rangle \\ &= \int_{\mathbb{R}^d} \varphi(x)f(x) d^d x, \end{aligned}$$

where we consider $\varphi(x)$ as a generalized function associated to φ . We would then consider the “derivative”

of the distribution φ by using integration by parts, so that

$$\begin{aligned}\int_{\mathbb{R}^d} \partial^\alpha \varphi(x) f(x) d^d x &= (-1)^{|\alpha|} \int_{\mathbb{R}^d} \varphi(x) \partial^\alpha f(x) d^d x \\ &= (-1)^{|\alpha|} \langle \partial^\alpha f, \varphi \rangle \\ &=: \langle f, \partial^\alpha(\varphi) \rangle.\end{aligned}$$